

Lecture 1: Evolution of Computers & Number Systems

*Before understanding Computer Networks, it's important to understand **how computers evolved** and **why binary numbers are used**.*

Why Did Humans Need Computers?

Initially, humans lived a simple, repetitive life:

Eat → Sleep → Repeat

As humans started domesticating animals (cows, goats, buffaloes, etc.), a new problem appeared:

How is an animal identified as missing?

This created the need for a way to **count** things — and counting is where the entire story of computing begins.

Tally Marks (First Counting System)

The earliest counting method was **Tally Marks**.

Working:

- ♦ One stone represented one animal.
- ♦ Every time an animal left for grazing, one stone was removed.
- ♦ When all animals returned, the remaining stones indicated missing animals.

1 Stone = 1 Animal

Problem:

This method works only for a small number of animals. Counting 100 or 1000 animals using stones quickly becomes impractical — there is no compact way to represent large quantities.

Birth of Number Systems

To solve this problem, humans invented **Number Systems**.

A Number System is a method of representing numbers using a fixed set of digits (symbols).

Instead of one stone per unit, a number system lets a small set of symbols represent any quantity, however large — this is the core idea that makes counting scalable.

Base-60 Number System

One of the earliest structured number systems.

Base = 60
60 unique symbols

Problem:

- ✦ Difficult to learn (60 distinct symbols to memorize)
- ✦ Difficult to perform calculations
- ✦ Not convenient for daily use

Decimal Number System (Base-10)

Developed in India, and eventually adopted worldwide.

It contains **10 unique digits**:

0 1 2 3 4 5 6 7 8 9

Why Base-10?

- ✦ Humans have **10 fingers**.
- ✦ Counting became easier and faster, since people could count naturally on their hands.

Example:

7 + 4

7 → 8 → 9 → 10 → 11

Answer = 11

Why Were Computers Invented?

During the **1800s**, trade and business increased rapidly, and large-scale calculations became difficult and error-prone for humans to perform manually.

Computers were developed to perform calculations (compute).

Computer comes from the word **Compute**, meaning **to calculate**. The earliest "computers" were not electronic machines at all — they were built to solve arithmetic-heavy problems faster and more reliably than people could.

Early Computers

The earliest computers were:

- ♦ Huge in size (occupying entire rooms)
- ♦ Mechanical, made of gears and moving parts
- ♦ Human-operated
- ♦ Very accurate
- ♦ Extremely slow compared to today's standards

Invention of Transistors

Around the **1950s**, **Transistors** replaced bulky mechanical and vacuum-tube components.

A transistor has only **two stable states**:

OFF → 0
ON → 1

This simple two-state behavior became the foundation of modern computing — every digital circuit today is built from millions (or billions) of transistors switching between these two states.

Binary Number System

Binary uses only **two digits**:

0
1

Hence:

Base = 2

Binary Addition Rules

A	B	RESULT
0	0	0
0	1	1
1	0	1
1	1	10

Decimal to Binary Conversion

Repeatedly divide by **2** and read the remainders from **bottom to top**.

Example: Convert **27** to Binary.

```
27 ÷ 2 = 13   Remainder 1
13 ÷ 2 = 6    Remainder 1
6 ÷ 2 = 3     Remainder 0
3 ÷ 2 = 1     Remainder 1
1 ÷ 2 = 0     Remainder 1
```

Answer = 11011_2

Binary to Decimal Conversion

Multiply each bit by its corresponding **power of 2**, then sum the results.

```
110112

= (1×24) + (1×23) + (0×22) + (1×21) + (1×20)
= 16 + 8 + 0 + 2 + 1
= 2710
```

Octal Number System

- ♦ Base = 8
- ♦ Digits = 0–7

Decimal to Octal Conversion

Repeatedly divide by **8** and read the remainders from **bottom to top**.

Example: Convert **37** to Octal.

$$37 \div 8 = 4 \quad \text{Remainder } 5$$
$$4 \div 8 = 0 \quad \text{Remainder } 4$$

Answer = 45_8

Octal to Decimal Conversion

Multiply each digit by its corresponding **power of 8**, then sum the results.

$$45_8$$
$$= (4 \times 8^1) + (5 \times 8^0)$$
$$= 32 + 5$$
$$= 37_{10}$$

*Octal is a convenient shorthand for binary, since each octal digit represents exactly **3 binary bits** ($2^3 = 8$).*

Hexadecimal Number System

- ♦ Base = 16
- ♦ Digits = 0–9, A–F

SYMBOL	DECIMAL
A	10
B	11
C	12
D	13
E	14
F	15

Decimal to Hexadecimal Conversion

Repeatedly divide by **16** and read the remainders from **bottom to top**.

Example: Convert **38** to Hexadecimal.

$$\begin{array}{ll} 38 \div 16 = 2 & \text{Remainder } 6 \\ 2 \div 16 = 0 & \text{Remainder } 2 \end{array}$$

$$\text{Answer} = 26_{16}$$

Hexadecimal to Decimal Conversion

Multiply each digit by its corresponding **power of 16**, then sum the results.

$$26_{16}$$

$$= (2 \times 16^1) + (6 \times 16^0)$$

$$= 32 + 6$$

$$= 38_{10}$$

*Hexadecimal is widely used in computing (memory addresses, color codes, MAC addresses) because each hex digit represents exactly **4 binary bits** ($2^4 = 16$), making it a compact way to write binary values.*

Why Do Computers Store Data in Binary?

Transistors can only be **ON** or **OFF**:

$$\text{OFF} = 0$$

$$\text{ON} = 1$$

Example:

$$13_{10} = 1101_2$$

Binary is the most natural way for computers to store data because it directly maps to the **ON/OFF** states of transistors — there is no simpler or more reliable state a circuit can hold.

Moore's Law

*The number of transistors on an integrated circuit doubles approximately every **2 years**.*

Impact:

- ♦ Faster computers
- ♦ Smaller devices
- ♦ More memory
- ♦ Lower cost per transistor

This exponential growth is why computing power has increased so dramatically over the past few decades, while devices have gotten smaller and cheaper.

Early Data Transfer

Initially, computers could not communicate directly with one another. There was:

- ✦ No Floppy Disk
- ✦ No Pen Drive
- ✦ No Internet

Data was transferred physically using **Punch Cards**.

Punch Card

A **Punch Card** was a paper sheet containing **rows and columns**, where each position represented one bit.

```
Hole    → 1
No Hole → 0
```

Working:

1. Data was punched manually onto the sheet.
2. A human carried the sheet to another location.
3. Another person inserted it into a Punch Card Reader.
4. The computer reconstructed the binary data from the pattern of holes.

Limitations:

- ✦ Slow data transfer
- ✦ Card damage could destroy data
- ✦ Human involvement required at every step
- ✦ Missing or incorrect punch could cause **data loss**

These limitations in early data transfer eventually led to the development of **Computer Networks**, which solved the problem of moving data between machines reliably and automatically.

Summary

- ♦ Humans first used Tally Marks to count objects, but this method became impractical for large quantities.
- ♦ The invention of Number Systems made counting and calculations more efficient and scalable.
- ♦ The Decimal Number System (Base-10) became the global standard because it is simple and naturally suited to human counting (10 fingers).
- ♦ Computers were invented in the 1800s to perform large and complex calculations quickly and accurately.
- ♦ The invention of transistors revolutionized computers by introducing two stable states: ON (1) and OFF (0).
- ♦ Binary (Base-2) is the fundamental language of computers because it directly matches transistor states.
- ♦ Decimal, Binary, Octal, and Hexadecimal are different ways of representing the same numerical values, each suited to different use cases.
- ♦ Moore's Law explains the rapid growth in computing power through the increasing number of transistors on integrated circuits.
- ♦ Early computers exchanged data using Punch Cards, which were slow, manual, and unreliable.
- ♦ The limitations of early data transfer methods directly led to the development of Computer Networks for faster and more reliable communication.